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Machine Learning

“TECHNOLOGY Is Creating NEW ISSUES That Would Require HUMANS TO ADDRESS THEM”

With new technologies like artificial intelligence and machine learning conquering the world, professionals need to acquire requisite skills and knowledge to advance further in their career. Nidhi Arora, executive editor, EFY magazine, in conversation with Renjith Pt, vice president, Livewire India, discusses areas of opportunities and key to success in this field

Q. What are the machine learning opportunities across industries?

A. According to marketsandmarkets, jobs market in machine learning is expected to grow annually at 44.1 per cent from US\$ 1.03 billion in 2016 to US\$ 8.81 billion by 2022. Organisations worldwide are adopting machine learning solutions to enhance customer experience and gain a competitive edge in their business. Technological advancement and huge data generation/storage are some of the market drivers.

In BFSI sector, machine learning is used for fraud and risk management, investment prediction, customer segmentation, digital assistance, compliance management and credit underwriting. In retail sector, it is used for inventory planning, segmentation and targeting, customisation management, recommendation engines and cross-channel marketing. Healthcare and life sciences sector uses machine learning for disease identification and diagnosis, image analytics, drug discovery/manufacturing, personalised treatment, clinical trial research and epidemic outbreak prediction. Manufacturing sector applications include predictive maintenance, demand forecasting, revenue estimation, supply chain management and root cause analysis. Telecommunication sector uses machine learning for customer analytics, network optimisation, network security, digital

assistance and marketing campaign analytics. Energy and utilities sector uses it for renewable energy management, power/energy usage analytics, smart grid management, seismic data processing and carbon emission. Government and defence sector applications include threat intelligence, autonomous defense system, sustainability and operational analytics.

Q. Does that mean loss of jobs for humans?

A. A McKinsey report states, “As ever more of the analogue world gets digitised, our ability to learn from data by developing and testing algorithms will only become more important for what is now seen as traditional businesses.”

Technologies such as machine learning, big data, artificial intelligence and automation are increasingly shaping the future of IT jobs. Repetitive and tedious jobs will get automated, but machines will always need humans. Technology is creating new issues that would require humans to address them. Also, human imagination or ingenuity can never be undermined.

Q. What are the challenges coming along?

A. Automation is stealing mid-level jobs. Moreover, machine learning and artificial intelligence are data hungry. In order to get more and more

data, we are now venturing into the privacy of individuals. Individuals should be made aware of rules and regulations on data collection and sharing. They should become more sensible on what they share or post on social media.

Q. What are the requisite skills to succeed in this space?

A. A few of the important skills for machine learning professionals include computer science fundamentals and programming skills, programming languages like R and Python, Minitab and Tableau tools, probability and statistics, data modeling and evaluation, software engineering's best practices and storytelling capabilities. The demand for data scientists is growing and it is projected to exceed supply by more than 50 per cent this year.

Q. What's the eligibility to enroll for machine learning courses?

A. We recommend this training for developers who want to become data scientists, analytics team managers, business analysts, information architects and analytics professionals who want to explore machine learning or artificial intelligence. Graduates looking to build a career as well as experienced professionals who want to gain more insights in their field of work can also opt for these courses. **EFY**